

# Product Carbon Footprint Study Report

---

Product Name: 1 tonne of cement

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

## I. General Information

**Producer Name:** [Information provided by the user]

**Address:** [Information provided by the user]

**Legal Representative:** [Information provided by the user]

**Product Name:** 1 tonne of cement

**Product Function:** [Information provided by the user]

**Based on the Standard:** IPCC 2013 GWP 100a

## II. Quantitative Purpose

[Information provided by the user]

## III. Scope of Quantification

### 1. Functional or Declaratory Units

Functional or declarative units in 1 kg cement.

### 2. System Boundary

The Life Cycle Assessment (LCA) of the traditional cement production process includes the following stages: the extraction of the primary raw materials (limestone and marlstone), the preparation of the raw meal with necessary additives, and the clinker production process, in Swend, in 2023.

### 3. Cut-off Criteria

[Information provided by the user]

### 4. Time Scale

[Information provided by the user]

## IV. Inventory Analysis

### 1. Description of Data Sources

**Primary Data:** [Information provided by the user]

**Secondary Data:** ecoinvent

### 2. Principles and Procedures for Allocation

**Basis of Allocation:** [Information provided by the user]

**Allocation Procedure:** [Information provided by the user]

**Specific Allocations:** [Information provided by the user]

### 3. Data Quality Evaluation (Optional)

[Information provided by the user]

## V. Impact Assessment

### 1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

### 2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of 1 tonne of cement produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1.005 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

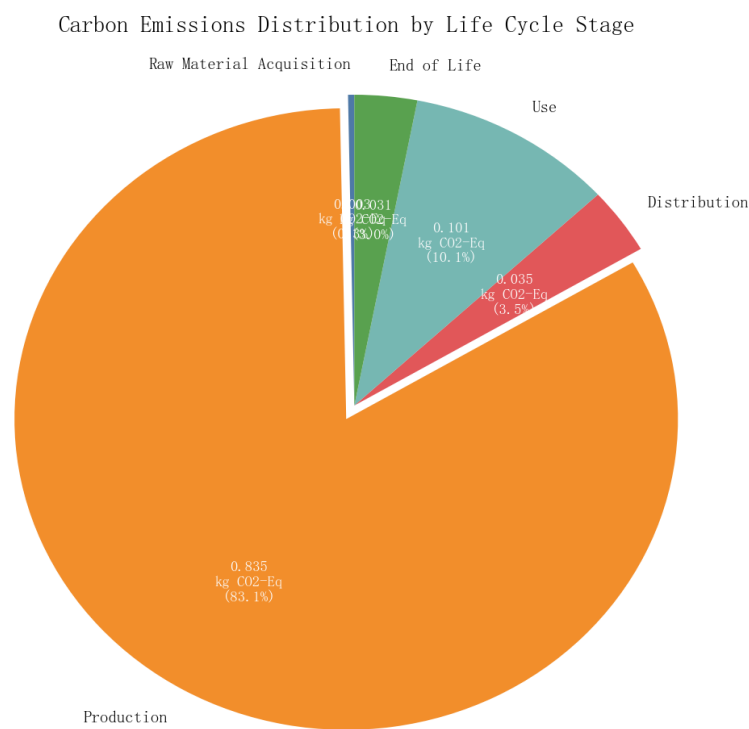


Table 1: Carbon Emissions by Life Cycle Stage

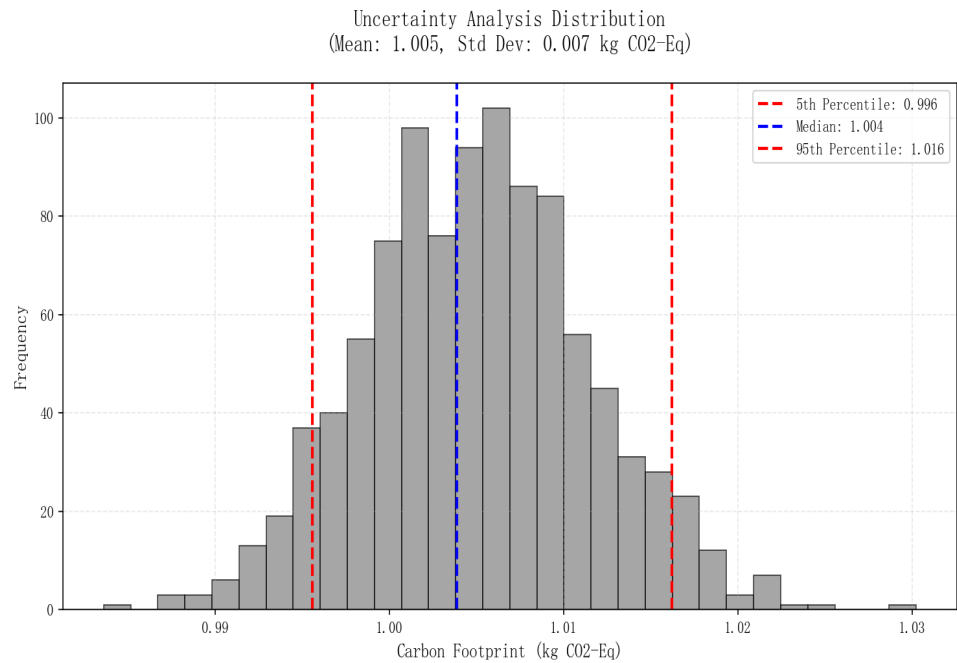
| Life Cycle Stage         | Carbon Footprint (kg CO2-Eq) | Percentage (%) |
|--------------------------|------------------------------|----------------|
| Raw Material Acquisition | 0.003                        | 0.30           |
| Production               | 0.835                        | 83.11          |
| Distribution             | 0.035                        | 3.47           |

|              |              |               |
|--------------|--------------|---------------|
| Use          | 0.101        | 10.08         |
| End of Life  | 0.031        | 3.05          |
| <b>Total</b> | <b>1.005</b> | <b>100.00</b> |

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



| Statistical Metric            | Value (kg CO2-Eq) |
|-------------------------------|-------------------|
| Mean                          | 1.005             |
| Standard Deviation            | 0.007             |
| Coefficient of Variation (CV) | 0.65%             |
| Median                        | 1.004             |
| 5th Percentile                | 0.996             |
| 95th Percentile               | 1.016             |

## VI. Interpretation of Results

### 1. Description of Results

The carbon footprint of 1 tonne of cement totals 1.005 kg CO<sub>2</sub>-Eq, representing a notably low emission intensity relative to conventional Portland cement benchmarks. This value reflects an overall efficient lifecycle profile, with emissions distributed across five stages in a manner that highlights process-related hotspots rather than raw material extraction or end-of-life management.

Lifecycle contributions reveal a clear dominance by the production stage at 0.835 kg CO<sub>2</sub>-Eq (83.1%), followed by the use phase at 0.101 kg CO<sub>2</sub>-Eq (10.1%). Distribution, end-of-life, and raw material stages contribute modestly at 3.5%, 3.0%, and 0.3% respectively. This pattern indicates that thermal energy consumption, clinker calcination processes, and associated electricity use within production constitute the primary emission drivers, while downstream and upstream activities remain secondary.

Major emission sources are therefore concentrated in manufacturing operations, underscoring the need for targeted interventions in kiln efficiency and fuel substitution to achieve meaningful reductions.

### 2. Statement of Assumptions and Limitations

This assessment assumes the application of generic secondary datasets for background processes, average European electricity mixes, and standard transport distances without site-specific primary measurements. It further presumes a functional unit equivalence based on typical cement performance characteristics and a 100-year global warming potential horizon consistent with IPCC AR6 factors.

Data limitations include reliance on industry-average rather than supplier-specific inventories, potential temporal mismatches in emission factors, and incomplete coverage of regional variations in raw material sourcing. Quality constraints may introduce uncertainty in the production-stage allocation, particularly regarding allocation methods for co-products and the representativeness of technology assumptions.

System boundaries encompass a cradle-to-grave scope covering raw material extraction through end-of-life treatment but exclude capital goods, employee commuting, and indirect land-use changes. The assessment does not extend to biogenic carbon accounting or potential rebound effects associated with product use.

### 3. Suggestions for Improvement

- Implement waste heat recovery systems and alternative low-carbon fuels (e.g., biomass or hydrogen blends) in the production stage to directly address the 83.1% contribution.
- Optimize clinker substitution with supplementary cementitious materials such as fly ash or calcined clay to lower process emissions without compromising product performance.
- Shift distribution logistics toward rail or electric vehicles and reduce average transport distances through localized sourcing to cut the 3.5% distribution share.

- Enhance primary data collection via supplier-specific energy audits and real-time monitoring to improve accuracy of the dominant production inventory.
- Explore carbon capture and storage integration at the kiln level as a long-term pathway for substantial absolute reductions in manufacturing emissions.